

CLASS 08 / OCT 13 / U05

ATTRACTORS

P3 CONTINUES — THE RULE STOPS COMING FROM THE INDEX

P3b DUE

P3c + P3d ISSUED TODAY

AN ATTRACTOR IS A NUMBER

For every point: how far is it from something? Use that number to drive a property.

THE WHOLE IDEA

```
for each point p:  
    d = distance(p, a)  
    value = f(d)
```

Everything else this week
is a choice of f.

MEASURE IT FIRST

$$d = \sqrt{(px - ax)^2 + (py - ay)^2}$$

Pythagoras, once per point. The answer is in model units.

Index told you where in the sequence. Distance tells you how far from something outside the collection.

LAST WEEK YOU BUILT THE POSITIONS. THIS WEEK YOU MEASURE THEM.

READ THE FIELD BEFORE YOU CHANGE ANYTHING

Distance is a value at every point. Look at it as grey before it becomes geometry.

01 MEASURE

Every point on a flat 2D grid gets a number: how far to the attractor.

02 DISPLAY

Show those numbers as grey.
This is a drawing already.

03 REMAP

Push that range into the range your property needs.

04 DRIVE

Only now does anything change. The field was finished first.

THE FIELD, READ AS A GREY VALUE ACROSS THE PLANE · NEAR → FAR



IF THE GREY IMAGE IS NOT ALREADY A GOOD DRAWING, THE LINES WILL NOT SAVE IT.

P3c requires the grayscale field, with a legend, beside the line drawing. That order is the assignment.

REMAP, AND SAY WHERE YOU CLAMPED

Distance comes out in model units. Your property needs a specific range.

THE OPERATION

```
t    = (d - dmin) / (dmax - dmin)
t    = clamp(t, 0, 1)
out  = outmin + t × (outmax - outmin)

d    0 ... 8.4 units    ← input
out  2 ... 12 mm length ← output
```

SUBTRACT. DIVIDE. MULTIPLY. ADD.

Nothing here is new. It is the arithmetic of a transformation, applied to a distance instead of a coordinate.

Clamping is a decision, not a safety net. Decide what happens past the end of your range and say so.

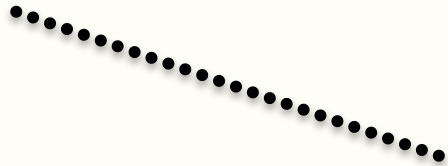
P3c REQUIRES INPUT AND OUTPUT RANGES, CLAMPING AND FALLOFF LABELLED ON THE SHEET.

FALLOFF IS THE FUNCTION

f is your decision. It is where the graphic quality actually comes from.

LINEAR

$$v = 1 - t$$



t (remapped distance) →

Even change all the way across.

INVERSE

$$v = 1 / (1 + k \cdot t)$$

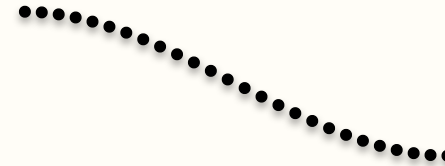


t (remapped distance) →

Sharp near the attractor, flat far off.

SMOOTH

eased



t (remapped distance) →

Soft at both ends. Often the best-looking.

STEP

on / off at a cut



t (remapped distance) →

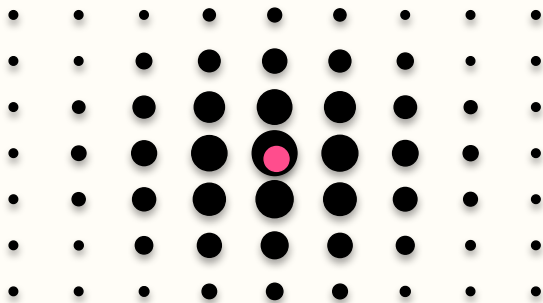
No gradient at all. Two populations.

Two students with the same attractor and different falloff produce completely different drawings. This is a design decision, not a setting.

THREE KINDS

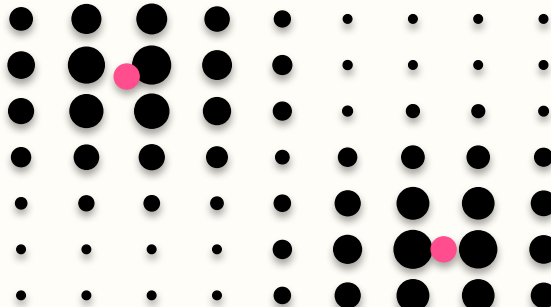
Same arithmetic. What changes is what you measure the distance to.

ONE POINT



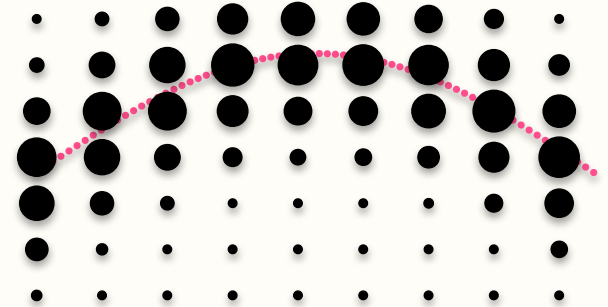
$$d = \text{distance}(p, a)$$

NEAREST OF SEVERAL



$$d = \min(\text{distance}(p, a_i))$$

NEAREST ON A CURVE



$$d = \text{distance}(p, \text{closest}(c, p))$$

Whatever you measure to, the output is the same thing: one number per point. P3c compares all three on the same grid, with the same mapping.

THE SAME THING, WRITTEN OUT

Measure, remap, clamp, shape the falloff, then place. In that order.

```
for p in points:
    d = distance(p, attractor)          # or min(...) over several
    t = (d - d_min) / (d_max - d_min)   # remap
    t = clamp(t, 0, 1)
    t = falloff(t)                     # linear, inverse, smooth, step
    size = out_min + t * (out_max - out_min)
    place(component, p, scale=size)
```

WHAT IT MAPS ONTO

distance
external condition
field
remap
clamp
falloff
nearest

LINE 2 IS THE ONLY LINE THAT CHANGES BETWEEN THE THREE ATTRACTOR TYPES.

ISSUED TODAY

P3c — ATTRACTOR: 2D FIELD PLOTTER DRAWING

DUE WEEK 10 — Oct 27 / 7%

Take the field off the screen. Sample it on a flat grid, turn every value into a line, and let a pen plotter draw it. Same grid, same mapping, three attractors: the drawing shows what the attractor alone changes.

FIELD

GREY FIRST

The grayscale field, with a legend, before any line.

LINES

ONE MAPPING

Length, rotation, spacing or hatch density, driven by distance.

PLOT

SVG TO PAPER

Export at plotted size, test small, then plot the sheet.

REQUIREMENTS — P3c

Visual documentation 20% · computational understanding 20% · technical execution 15% · fabrication quality 30% · experimentation 5% · requirements and identity 10%.

01

Build a 2D attractor field on a flat grid and translate it into lines a pen plotter can draw — vary line length, rotation, spacing or hatch density with the measured distance.

02

Compare single-point, multiple-point and curve attractors on the same grid, with the same mapping, so the difference comes from the attractor alone.

03

Show the grayscale field with a legend beside the line drawing. Label input and output ranges, clamping and falloff.

04

Export an SVG at the plotted size, test a small plot, then plot the final drawing at 17 × 11 inches (or the confirmed plotter size) with one pen weight or a declared set of pens.

05

Submit the plotted drawing, the SVG, a photograph or scan of the plot and the editable graph. Move an attractor or change the mapping live at midterm.

CHECKPOINT · WEEK 9 — Oct 20 — Bring a grayscale field, the line translation and a small test plot.

ALSO ISSUED TODAY

P3d — ATTRACTOR: SITE ANALYSIS WITH MIXTLI

DUE WEEK 10 — Oct 27 / 7%

Pick a real site and analyze it with Mixtli, the Blender add-on the instructor provides. Read its output as a field — the same distance-and-remap logic as P3c — and use it to make one spatial decision.

SITE

RECORD IT

Source, units, scale and orientation, before any analysis.

ANALYZE

ONE COMPARISON

One analysis with a legend and units, and one comparison.

DECIDE

YOUR READING

The add-on computes. The question and the reading are yours.

REQUIREMENTS — P3d

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Pick a site you can document and state why you chose it. Record the source of the site model or data, its units, scale and orientation.

02

Analyze the site with the Mixtli add-on for Blender, provided by the instructor. Record the analysis type, inputs and settings.

03

Make one 17 × 11 inch sheet with an annotated site plan or base view, the mapped analysis with a legend and units, and one comparison of settings, times or conditions.

04

Explain what the output measures, how it reads as a field — the same distance-and-remap logic as P3c — and how it informs one spatial decision on the site.

05

Submit the working file and dependencies. The add-on does the computation; the question, comparison and interpretation are yours.

CHECKPOINT · WEEK 9 — Oct 20 — Bring your chosen site, one Mixtli analysis with a legend and a stated analysis question.

NOW — YOUR FIRST FIELD

Rest of the class: measure, look, remap, and only then change anything.

✓

A flat 2D grid, and distance to one point measured at every sample

✓

The field displayed as grey, with a legend, before anything changes

✓

Input and output ranges written down — actual numbers, not guesses

✓

Clamp and falloff chosen, and you can say why

✓

One point swapped for several, then for a curve — same mapping throughout

MOVE THE ATTRACTOR. IF YOU CANNOT PREDICT WHAT HAPPENS, STOP AND READ THE FIELD.

BEFORE NEXT CLASS

NEXT

Week 9 — Oct 20: plotter workflow — field to lines to SVG to test plot — and a Mixtli working review of your site.

P3c

Checkpoint next class. Bring a grayscale field, the line translation and a small test plot.

P3d

Checkpoint next class. Bring your chosen site, one Mixtli analysis with a legend and a stated analysis question.

PLAN

The midterm is Week 10 — Oct 27. Projects 1-3 are presented, printed and plotted. Do not leave prints or plots to the last week.